ARM PrimeCell

Colour LCD Controller (PL110 r1p2)

Release Note

PL110 RELN 0000 E

Open Access

ARM PrimeCell Colour LCD Controller (PL110) Release Note

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General suggestion for additions and improvements are also welcome.

Feedback on the ARM Primecell Colour LCD Controller

If you have any comments or suggestions about this product, please contact your supplier giving:

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Change history

A	5th August, 1999	First release.
B	24th September, 1999	Beta 2.0 Release
C	29th September, 1999	Release 1.0
D	7th December, 2000	Release 1.1
E	28th March, 2003	Release r1p2

Contents

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL110 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL110-MN-23002-r1p2-00ltd0
Synthesizable Verilog RTL	N/A	PL110-MN-22002-r1p2-00ltd0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB18 cell library)	N/A	PL110-MN-01002-r1p2-00ltd0
Testbench - Functional Verification, VHDL	N/A	PL110-MN-23012-r1p2-00ltd0
Testbench - Integration test, VHDL	N/A	PL110-MN-23010-r1p2-00ltd0
Testbench - Functional Verification, Verilog	N/A	PL110-MN-22012-r1p2-00ltd0
Testbench - Integration test, Verilog	N/A	PL110-MN-22010-r1p2-00ltd0
Test vectors-BusTalk, functional verification	N/A	PL110-VE-01012-r1p2-00ltd0
Test vectors-BusTalk, integration test	N/A	PL110-VE-01010-r1p2-00ltd0
ARM Primecell Colour LCD Controller (PL110) Technical Reference Manual (PDF)	ARM DDI 0161 E	PL110-DA-03001-r1p2-00ltd0
ARM Primecell Colour LCD Controller (PL110) Design Manual (PDF)	PL110 DDES 0000 C	PL110-DC-02002-r1p2-00ltd0
ARM Primecell Colour LCD Controller (PL110) Integration Manual (PDF)	PL110 INTM 0000 C	PL110-DC-10002-r1p2-00ltd0
ARM Primecell Colour LCD Controller (PL110) Release Note (PDF)	PL110 RELN 0000 E	PL110-DC-06002-r1p2-00ltd1
ARM Primecell Colour LCD Controller (PL110) Errata Notice (PDF)	PL110 DEI 0000	PL110-DC-11001-r1p2-00ltd0
ARM Primecell Colour LCD Controller (PL110) Userguide (PDF)	PL110 USGD 0000 B	PL110-DC-08000-r1p2-00ltd0
The parts with numbers starting PL000 are delivered as a separate bundle PL000-BU-00004		
System Common and System Indirection Source Code	N/A	PL000-SW-02001-r9p0-00rel0
System Test Source Code	N/A	PL000-SW-02003-r9p0-00rel0
System Common Files Source Code	N/A	PL000-SW-02004-r9p0-00rel0
System Build Project Files	N/A	PL000-SW-02005-r9p0-00rel0
Interrupt Controller Object Code	N/A	PL001-SW-00000-r9p0-00rel0
Timer Object Code	N/A	PL002-SW-02001-r9p0-00rel0

Software Code Module	N/A	PL110-SW-02001-r1p2-00ltd0
Software Test Report	PL110_TREP_1000	PL110-DC-01001-r1p2-00ltd0
Software API Document	PL110_ISPC_1000	PL110-DC-02001- r1p2-00ltd0
System Structure and Integration Guide	PL000_DII_1000	PL000-DC-02000-r9p0-00rel0
System Common API Documents	PL000_ISPC_1001	PL000-DC-02001-r9p0-00rel0
System Indirection API Documents	PL000_ISPC_1002	PL000-DC-02001-r9p0-00rel0
System Test API Documents	PL000_ISPC_1003	PL000-DC-02003-r9p0-00rel0
System Common Files API Documents	PL000_ISPC_1004	PL000-DC-02004-r9p0-00rel0
Interrupt Controller API Documents	PL001_ISPC_1000	PL001-DC-02001-r9p0-00rel0

Table 1.1-1: ARM part numbers for ARM PrimeCell CLCDC PL110

Note Only the required deliverables will be supplied, as per the contractual agreement.

1.2 Reference Cell Library

A reference cell library (Avant! CB18 0.18μm standard cells) is supplied with this peripheral.
PLXXX_PrimeCell/PL000-BU-00005-0.00/Avanti/CB18_2000.3.tar.gz
PLXXX_PrimeCell/PL000-BU-00005-0.00/Avanti/CB18_2000.3.lst

1.3 Common driver bundle

A bundle containing drivers, common to all Primecells is also supplied.

This bundle is part PL000-BU-00004 and is supplied as a single tarred and zipped file,

PL000-BU-00004-r9p0-00rel0.tgz

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX compressed tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 87 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and encrypted file of following format:

E.g.

Company Name-PrimeCell-PL110–Bundled-arm-download-WWWW.tgz.pgp

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

The first three steps may have already been done, ie the shipment from ARM may have already been decrypted and unzipped. If this is the case then go to step 4, else start with step 1.

1. **Relocate the shipment file**

   Relocate the encrypted file (for example <file>.tgz.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space if PGP encrypted will need the relevant keys.

2. **Decrypt file**

   Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tgz

3. **Unzip file**

   Execute the UNIX unzip command to obtain the tar file <file>.tar:

   gunzip <file>.tgz

   where <file> is the name of the file supplied.

4. Extract tar files

To extract the PrimeCell, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

This will generate the following files

ARM_MANIFEST_XXXX.TXT – which lists the PrimeCell files delivered

ARM_DELIVERY_XXXX.TXT – which lists the ARM parts delivered

And will unpack the PrimeCell into the bundle directory PL110-BU-00000-r1p2-00ltd1

Beneath this bundle directory are 2 sub-directories

PL110-BU-00000-r1p2-00ltd1/PL110_VC contains the PrimeCell development environment

PL110-BU-00000-r1p2-00ltd0/PL110_SW contains software for the PrimeCell

The peripheral development environment PL110-BU-00000-r1p2-00ltd1/PL110_VC contains two further directory structures, a peripheral specific directory structure and a project global directory structure. The top level directory structures are shown in ***Figure 2-1.***

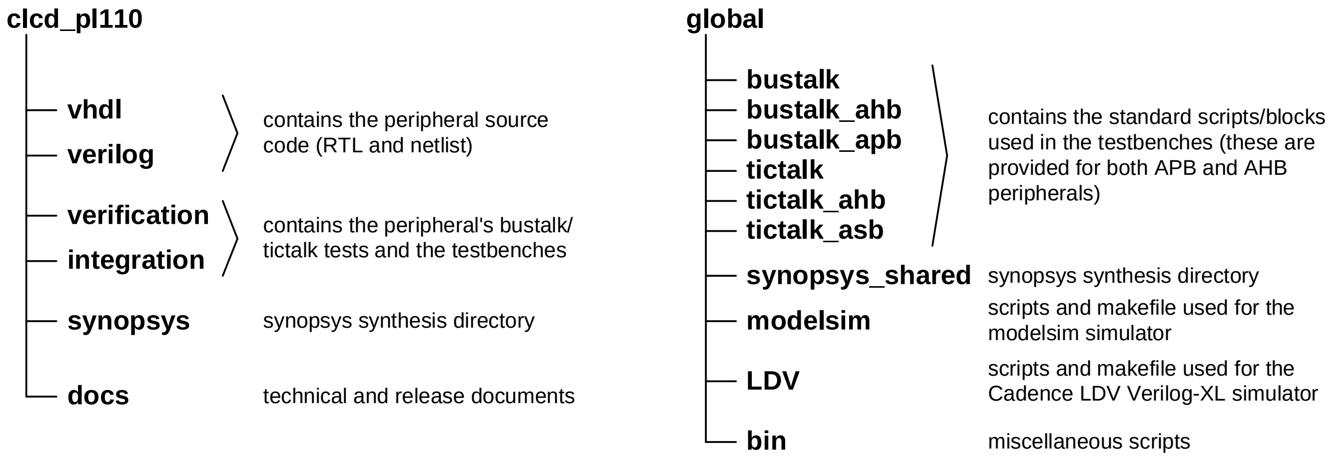

Figure 2-1. Peripheral Development Environment Directory Structure

2.3.2 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB18 1999.1 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PL110_PrimeCell/PL000-BU-00005-0.00/Avanti/CB18_2000.3.tar.gz .
```

Execute the UNIX uncompress and tar extract commands as below:

```
unzip CB18_2000.3.tar.gz
tar -xvf CB18_2000.3.tar
```

2.3.3 Installation of the Common Driver Bundle

A common software bundle PL000-BU-00004-r9p0-00rel0 is supplied with this delivery.

The files in the common driver bundle unzip and untar into a single directory PrimeCell_gen_sw.

The top level directory structures are shown in ***Figure 2-2.***

```
PrimeCell_gen_sw/
  ├── PL000_DII_1000.pdf
  ├── Documentation/
  └── Software/
        ├── apcommon/
        ├── apint/
        ├── apos/
        ├── aptest/
        ├── aptimer/
        └── build/
```

Figure 2-2 Common PrimeCell Driver Directory structure

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at http://www.adobe.com. Select “Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM Primecell Colour LCD Controller PL110:
The documents files are located in the CLCDC_PL110/docs/ directory.

ARM Primecell Colour LCD Controller PL110
Technical Reference Manual ARM DDI 0161

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM Primecell Colour LCD Controller PL110
Integration Manual PL110 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM Primecell Colour LCD Controller PL110
Design Manual PL110 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

ARM Primecell Colour LCD Controller PL110
Errata Notice PL110 DEI 0000

The errata notice describes details of any known issues with this ARM PrimeCell block and any workarounds for these issues. In most cases this document will be blank or may not be present.

ARM Primecell Colour LCD Controller PL110
User Guide PL110 USGD 0000

The user guide describes the instructions on performing simulations and synthesis for this ARM PrimeCell block.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release, it has not been implemented in production silicon.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM Primecell Colour LCD Controller PL110.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 2000.3 CB18 v1.0 (0.18 µm)	Avant! Passport 2000.3 CB18 v1.0 (0.18 µm)
Total cell area, without scan, NAND-2 equivalents	26631 (including palette RAM)	26726 (including palette RAM)
Net interconnect area, without scan, NAND-2 equivalents	6734 (including palette RAM)	6805 (including palette RAM)
Total cell area, scan inserted,NAND-2 equivalents	34320 (including palette RAM)	34466 (including palette RAM)
Net interconnect area, scan inserted,NAND-2 equivalents	9074 (including palette RAM)	9144 (including palette RAM)
Clock frequency used	166MHz	166MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100% (1870)	100% (1870)
Estimated scan fault coverage, collapsed.	99.35%	99.43%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2002.05-SP2	Synopsys Design Compiler and Test Compiler 2002.05-SP2
Simulation tools and versions	Model Technology Inc, ModelSim v5.5d	Model Technology Inc, ModelSim v5.5d Cadence Verilog-XL LDV3.2
Global Peripheral Development Environment Version	Release REL1v6	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running HCLK, CLCDCLK and nCLCDCLK clocks.

4.3.1 Synchronous clocks

HCLK	CLCDCLK, nCLCDCLK	Simulations
166 MHz (6ns period)	166 Mhz (6ns period)	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM Primecell Colour LCD Controller (PL110) Design Manual.

5 DIFFERENCES FROM PREVIOUS REVISIONS

5.1 Differences between REL 1v1 and r1p2-00ltd1

5.1.1 Major changes that have been made in this release

Palette Ram has changed from Asynchronous to Synchronous

Integration test registers have been added

Integration testbench has been added

PCD has been increased from 5 bits to 10 bits for the clock divider

PL110 r1p2 no longer supports a RAM based FIFO implementation by the instantiation of a Tpram block. Previous versions of the PL110, 1v0 and 1v1 did support both the register based and RAM based FIFO implementations. There remains legacy RTL within the design for the RAM based FIFO implementation, but such a change is not validated by the existing PL110 testbenches, and is not supported by ARM for this release.

5.1.2 The following Errata have been corrected in this release

3.1 Burst reads from registers return incorrect data
3.2 CLCP Can stop
3.3 Incorrect data masking in test vectors
3.4 Vector width mismatches
3.5 Signal missing from sensitivity list
3.6 Watermark Level Inconsistency

Please see the Errata Notice PL110 DEI 000 for more details

6 KNOWN ISSUES

At the time of release there is one known issue with release r1p2-00ltd1

Errata no. 4.1 Timing violations may occur in high-performance systems

A write to LCDTiming0-3 or LCDControl, followed immediately by a read from that register may result in the old value being read back, or a transient incorrect value being sampled. If sufficient time is allowed for the register to be updated in the CLCDCLK domain, and this value to propagate back to the HCLK domain, then the correct value will be seen.

This issue is believed to be of low severity, as the software is unlikely to access the peripheral in a way which will show this problem.

Any further issues found after the release of PL110 r1p2-00ltd1, will be documented in future releases of the PL110 Errata Notice PL110 DEI 0000

7 REVISION HISTORY

Date	Revision	Description
16-Jul-1999	ALP1v0	Pre-release of Alpha 1.0 version.
5-Aug-1999	BET1v0	Beta 1v0 Release
24-Sept-1999	BET2v0	Includes support for 24bpp and programmable FIFO watermark
29-Sept-1999	REL1v0	Release 1.0
07-Dec-2000	REL1v1	Release 1.1 (Maintenance Release)
28-Mar-2003	r1p2	Release r1p2 (Maintenance Release)